



# Fundamentals of Optics

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## Rules

- Final exam: 40%
- Mid-term exam: 30%
- Homework: 20%
- Attendance: 10%

## Reference books

**Optics, by Eugene Hecht**  
光学, 赵凯华、钟锡华

# CHAPTER 8

## INTERFERENCE

杨氏双缝干涉

&

迈克尔逊干涉

# Interference in daily life



**Optical interference** corresponds to the interaction of two or more lightwaves yielding a resultant irradiance that deviates from the sum of the component irradiances.

**Wavefront splitting (分波阵面):** portions of the primary wavefront are used either directly as sources to emit secondary waves or in conjunction with optical devices to produce virtual source of secondary waves. These secondary waves are then brought together to interference.

Example: Young's double slit experiment

**Amplitude splitting (分振幅) :** the primary wave itself is divided into two segments, which travel different paths before recombining and interfering.

Example: fringes of equal inclination

# 8.1 General Considerations

Principle of superposition:

The electric field intensity at a point in space:

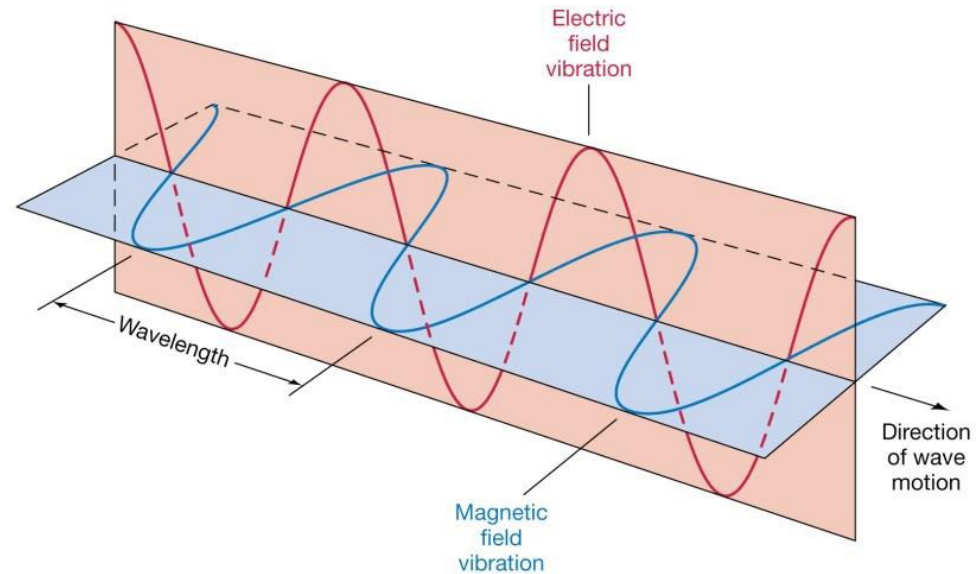
$$\vec{E} = \vec{E}_1 + \vec{E}_2 + \dots$$

The frequency of optical disturbance:

$$4.3 \times 10^{14} \text{ Hz to } 7.5 \times 10^{14} \text{ Hz}$$

The actual field is very difficult to detect, but the irradiance can be measured directly with sensors.

The irradiance: 
$$I = \epsilon v \left\langle \vec{E}^2 \right\rangle_T$$



Electromagnetic field

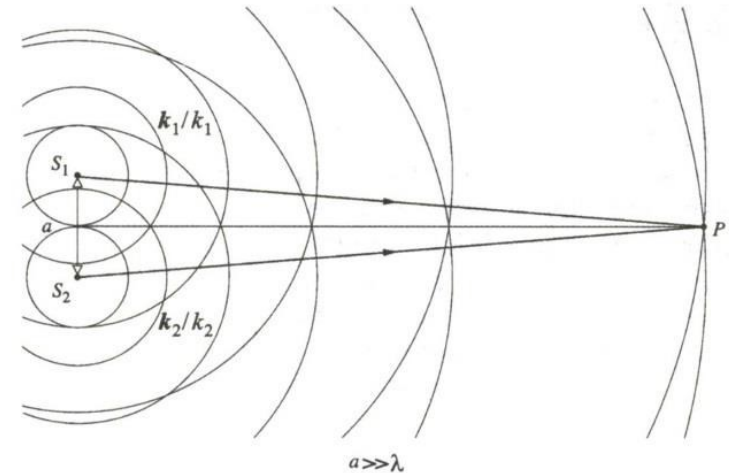
Case: consider two point sources emitting monochromatic waves of the same frequency in the same homogeneous medium;

- 1)  $a(\text{the separation}) \gg \lambda(\text{the wavelength})$ ;
- 2) point of observation P is far from the source.

Two linearly polarized waves:

$$\vec{E}_1 = \vec{E}_{01} \cos(\vec{k}_1 \cdot \vec{r} - \omega t + \epsilon_1)$$

$$\vec{E}_2 = \vec{E}_{02} \cos(\vec{k}_2 \cdot \vec{r} - \omega t + \epsilon_2)$$



Irradiance at P (neglect the constant terms):

$$I = \langle \vec{E}^2 \rangle_T = \langle \vec{E} \cdot \vec{E} \rangle_T = \langle (\vec{E}_1 + \vec{E}_2) \cdot (\vec{E}_1 + \vec{E}_2) \rangle_T$$

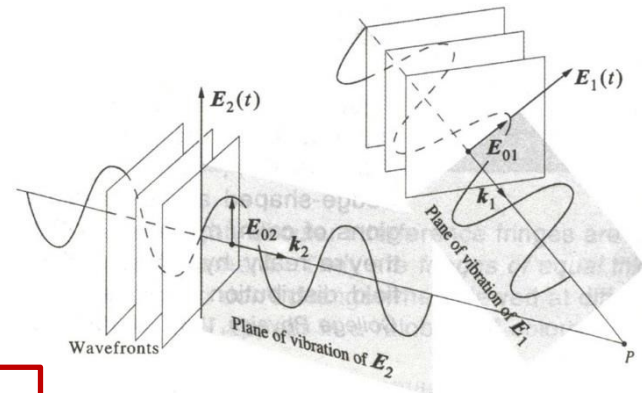
$$= \langle (\vec{E}_1^2 + \vec{E}_2^2 + 2\vec{E}_1 \cdot \vec{E}_2) \rangle_T = I_1 + I_2 + I_{12}$$

$$I_1 = \langle \vec{E}_1^2 \rangle_T$$

$$I_2 = \langle \vec{E}_2^2 \rangle_T$$

$$I_{12} = 2 \langle \vec{E}_1 \cdot \vec{E}_2 \rangle_T$$

Interference term



Interference term

$$I_{12} = 2 \left\langle \vec{E}_1 \cdot \vec{E}_2 \right\rangle_T = \vec{E}_{01} \cdot \vec{E}_{02} \cos \left( \vec{k}_1 \cdot \vec{r} - \vec{k}_2 \cdot \vec{r} + \epsilon_1 - \epsilon_2 \right) = \vec{E}_{01} \cdot \vec{E}_{02} \cos \delta$$

$\delta$ : phase difference arising from a combined path length and initial phase angle difference.

Case I: when  $\vec{E}_{01}$  and  $\vec{E}_{02}$  are perpendicular,  $I_{12} = 0$ ,  $I = I_1 + I_2$

Two such orthogonal polarized states will combine to yield an right-handed circularly polarized, left-handed circularly polarized, linearly polarized or elliptical polarized light (chapter 7), but the flux-density distribution will be unaltered.

Case II: when  $\vec{E}_{01}$  and  $\vec{E}_{02}$  are parallel

$$I_1 = \left\langle \vec{E}_1^2 \right\rangle_T = \frac{E_{01}^2}{2} \quad I_2 = \left\langle \vec{E}_2^2 \right\rangle_T = \frac{E_{02}^2}{2}$$

$$I_{12} = \vec{E}_{01} \cdot \vec{E}_{02} \cos \delta = 2\sqrt{I_1 I_2} \cos \delta$$

$$\text{Total irradiance: } I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \delta$$

Total constructive interference:  $\delta=0, \pm 2\pi, \pm 4\pi \dots$

$$I_{max} = I_1 + I_2 + 2\sqrt{I_1 I_2}$$

Constructive interference:  $0 < \cos \delta < 1$

$$I_1 + I_2 < I < I_{max}$$

When  $\cos \delta = 0$ ,  $I = I_1 + I_2$

Destructive interference:  $0 > \cos \delta > -1$

$$I_1 + I_2 > I > I_{min}$$

Total destructive interference:  $\delta=0, \pm 3\pi, \pm 5\pi \dots$

$$I_{min} = I_1 + I_2 - 2\sqrt{I_1 I_2}$$

Special case:  $E_{01} = E_{02}$ ,  
let  $I_1 = I_2 = I_0$

$$I = 2I_0(1 + \cos \delta)$$

$$= 4I_0 \cos^2 \delta / 2$$

$$I_{max} = 4I_0 \text{ and } I_{min} = 0$$



Spherical waves emitted by  $S_1$  and  $S_2$

$$\vec{E}_1(r_1, t) = \vec{E}_{01}(r_1) \exp[i(kr_1 - \omega t + \epsilon_1)]$$

$$\vec{E}_2(r_2, t) = \vec{E}_{02}(r_2) \exp[i(kr_2 - \omega t + \epsilon_2)]$$

$r_1, r_2$ : the radii of the spherical wavefronts overlapping at P

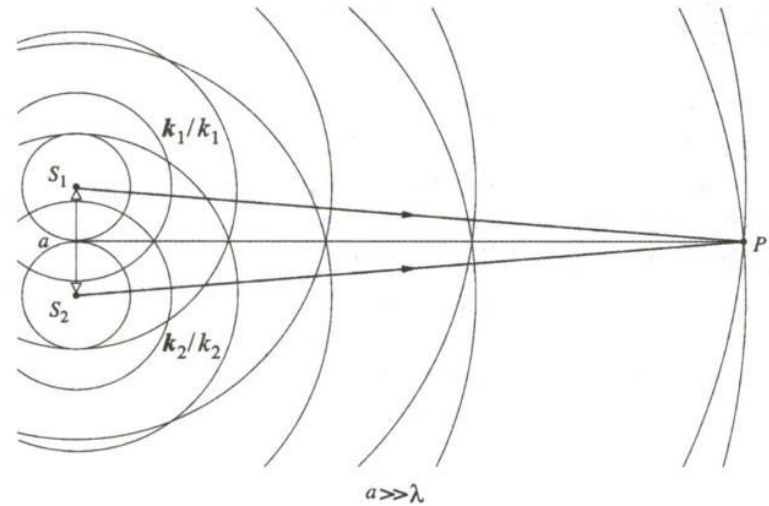
Irradiance for spherical waves:

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \delta$$

$$\delta = k(r_1 - r_2) + (\epsilon_1 - \epsilon_2)$$

When the separation ( $a$ ) is small in comparison with  $r_1$  and  $r_2$ , and the interference region is also small,  $\vec{E}_{01}, \vec{E}_{02}$  is considered independent of position, and assume that  $\vec{E}_{01} = \vec{E}_{02}$

$$I = 4I_0 \cos^2 \frac{1}{2} [k(r_1 - r_2) + (\epsilon_1 - \epsilon_2)]$$



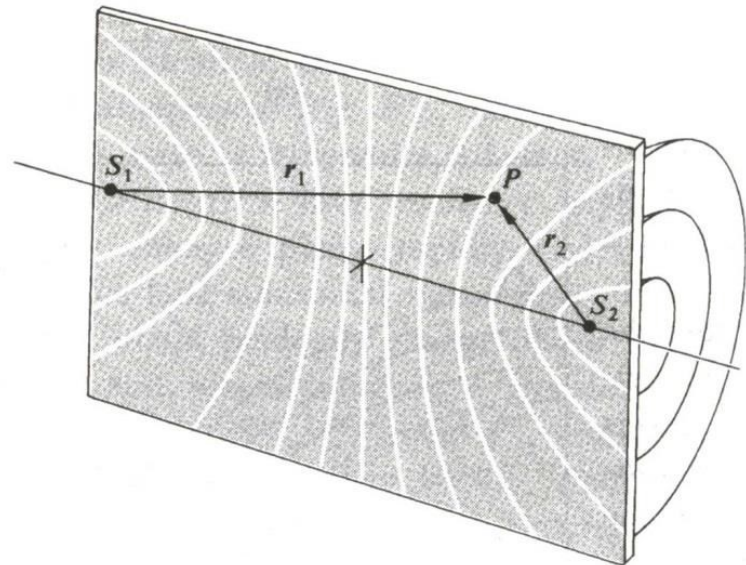
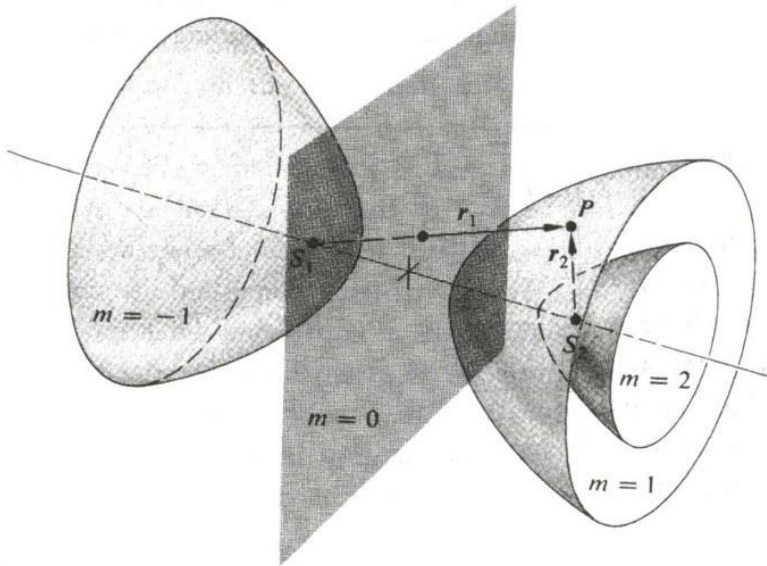
$$I = 4I_0 \cos^2 \frac{1}{2} [k(r_1 - r_2) + (\epsilon_1 - \epsilon_2)]$$

Irradiance maxima:  $\delta = 2\pi m; m = 0, \pm 1, \pm 2, \dots$

$$(r_1 - r_2) = \frac{[2\pi m + (\epsilon_1 - \epsilon_2)] \text{ In-phase } 2\pi m}{k} = m\lambda$$

Irradiance minima:  $\delta = (2m+1)\pi; m = 0, \pm 1, \pm 2, \dots$

$$(r_1 - r_2) = \frac{[(2m+1)\pi + (\epsilon_1 - \epsilon_2)] \text{ Out-phase } (2m+1)\pi}{k} = \left(m + \frac{1}{2}\right)\lambda$$



# 8.2 Conditions for Interference

## Condition I: The same frequency

The interference term for two waves with different frequency:

$$\begin{aligned} & 2 \cos(\omega_1 t + \varphi_1) \cos(\omega_2 t + \varphi_2) \\ &= \cos[(\omega_1 + \omega_2)t + \varphi_1 + \varphi_2] + \cos[(\omega_1 - \omega_2)t + \varphi_1 - \varphi_2] \end{aligned}$$

The  $I_{12}$  averages to zero during the detection interval.

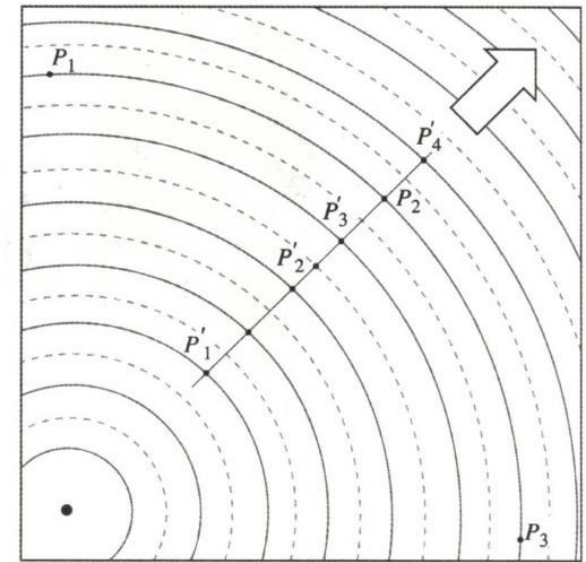
For white light:

The reds will interference with red, and the blues with blues.

## Condition II: coherence

Conventional quasimonochromatic sources produce light that is a mix of photon wavetrains.

Coherent time and length: average time interval and corresponding spatial extent during which the lightwave oscillates in a predictable way.

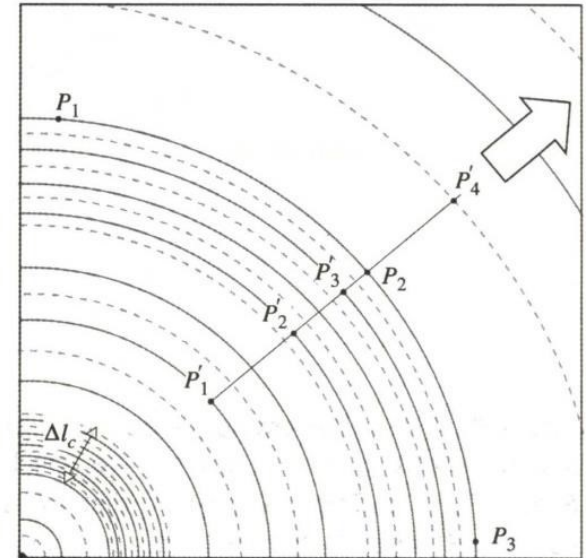


(a)

The waves display both forms of coherence perfectly

**Temporal coherence:** the interval over which the lightwave resembles a sinusoid before it randomly changes phase (about 10 ns).

**Spatial coherence:**  $P_1, P_2, P_3$  lie on the same wavefront in both cases, and the behaviors of waves at these points is completely correlated.



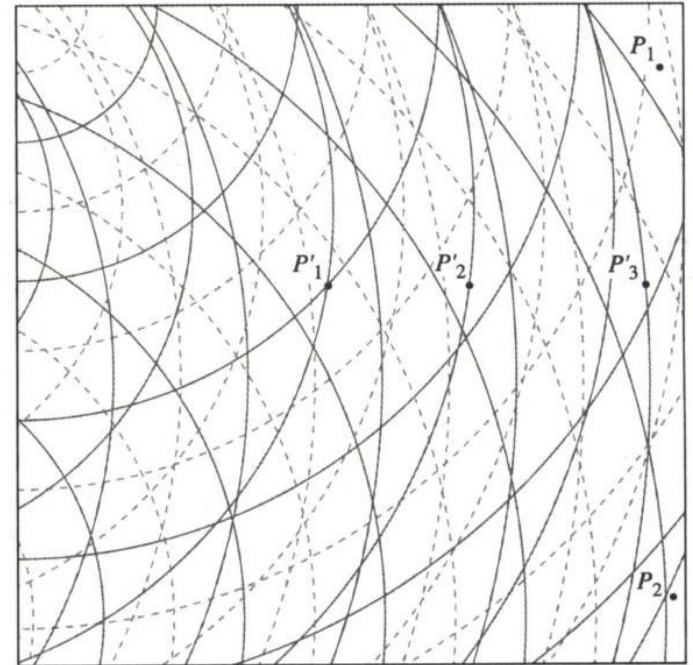
(b)

There is complete spatial coherence but only partial temporal coherence

Broad source (compose of many widely spaced point sources)

Each point source changes phase rapidly and randomly emitting 10-ns long sinusoidal wavetrains.

- ❑ The disturbance at  $P_1'$ ,  $P_2'$ , and  $P_3'$  is only correlated for a time less than 10ns;
- ❑ The lateral point such as  $P_1$  and  $P_2$  is completely uncorrelated depending on the size of the source.

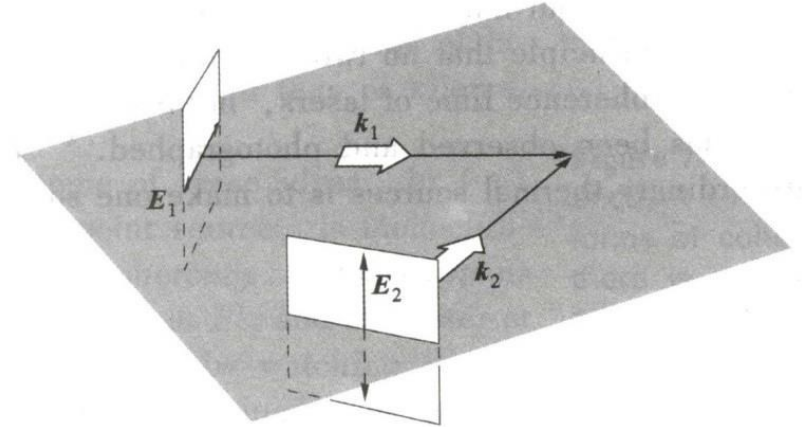
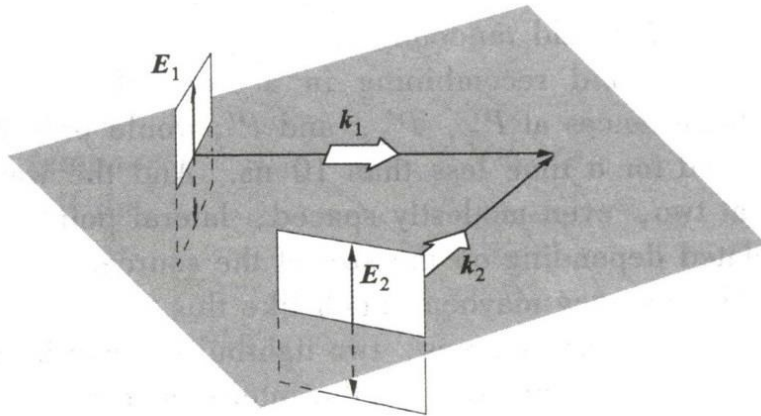


So it is impractical to observe interference for two sources, which maintain a constant relative phase for a time less than  $\Delta t$ .

Solution : one source serve to produce two coherent secondary sources.

Interference term

$$I_{12} = \vec{E}_{01} \cdot \vec{E}_{02} \cos \delta$$



Fresnel-Arago laws:

菲涅尔-阿拉戈定律

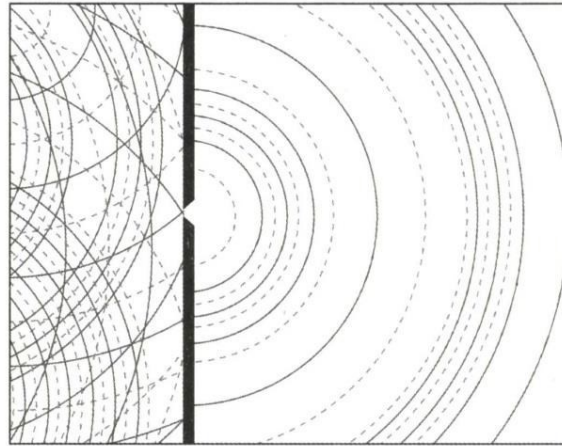
- ☐ Two orthogonal, coherent  $\wp$ -states cannot interfere in the sense that  $I_{12} = 0$  and no fringes result.
- ☐ Two parallel, coherent  $\wp$ -state will interfere in the same way as will natural light.
- ☐ The two constituent orthogonal  $\wp$ -states of natural light cannot interfere to form a readily observable fringe pattern even if rotated into alignment. Since these  $\wp$ -states are incoherent.

# 8.3 Wavefront-splitting Interferometers

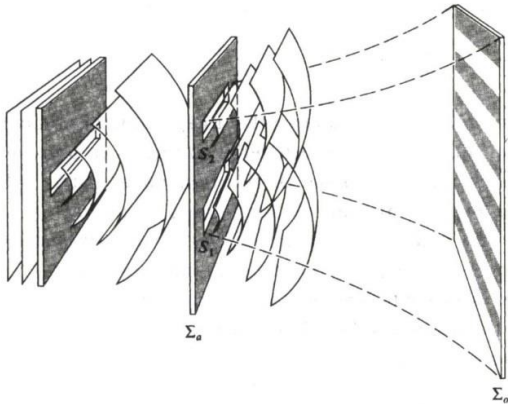
干涉计

Young's experiment

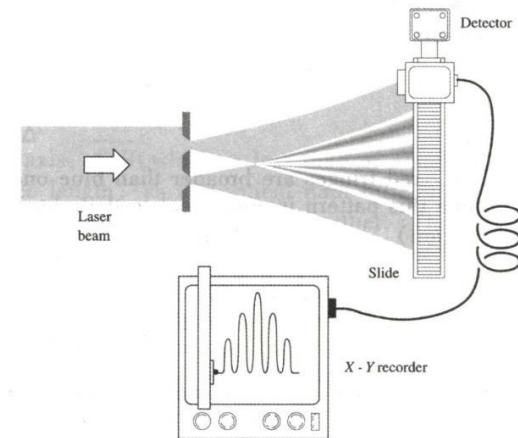
杨氏双缝干涉实验



Classic arrangement of  
Young's experiment



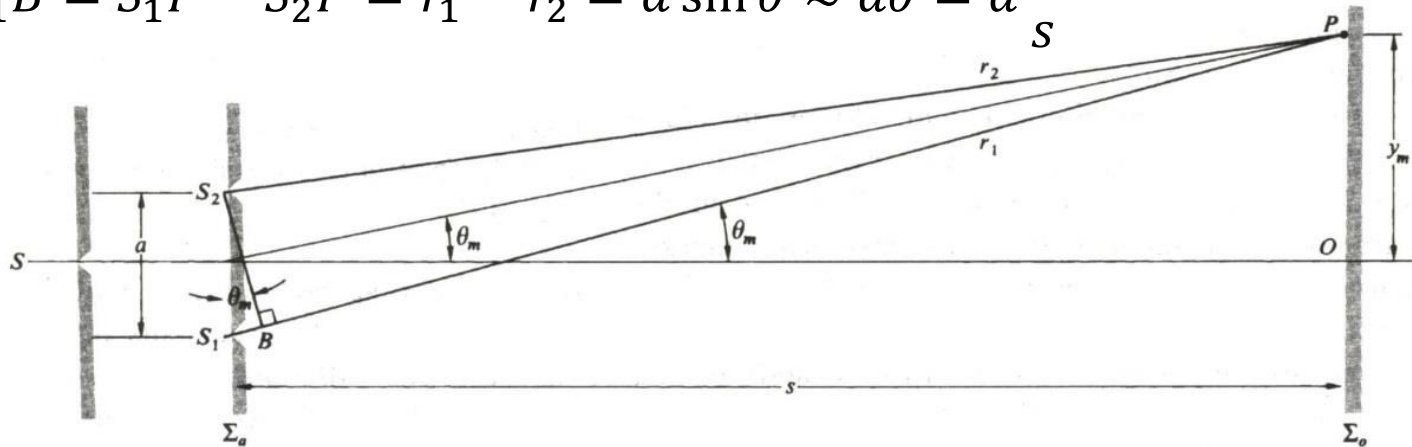
A modern version





The optical path difference

$$\overline{S_1B} = \overline{S_1P} - \overline{S_2P} = r_1 - r_2 = a \sin \theta \approx a\theta = a \frac{y}{s}$$



Constructive interference will occur when  $r_1 - r_2 = m\lambda$

The position of the m-th bright fringe on the screen (the maximum at 0 is the zeroth fringe)

$$y_m \approx \frac{s}{a} m\lambda$$

The angular position of the m-th bright fringe:  $\theta_m \approx \frac{y_m}{s} = \frac{m\lambda}{a}$

The spacing of the fringes on the screen:

$$\Delta y = y_{m+1} - y_m \approx \frac{s}{a} (m+1)\lambda - \frac{s}{a} m\lambda = \frac{s\lambda}{a}$$

Note: Red fringes are broader than blue ones.



Provided the two beams: coherence; equal irradiances

$$I = 4I_0 \cos^2 \frac{k(r_1 - r_2)}{2}$$

Phase difference:

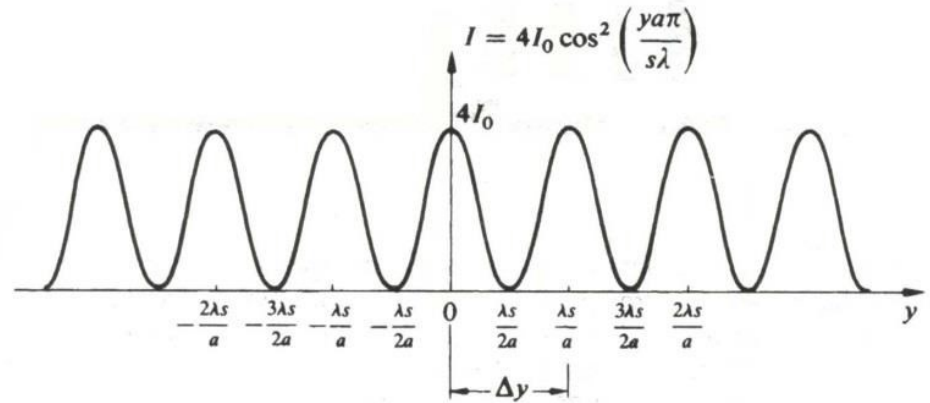
$$\delta = k(r_1 - r_2)$$

Optical path difference:

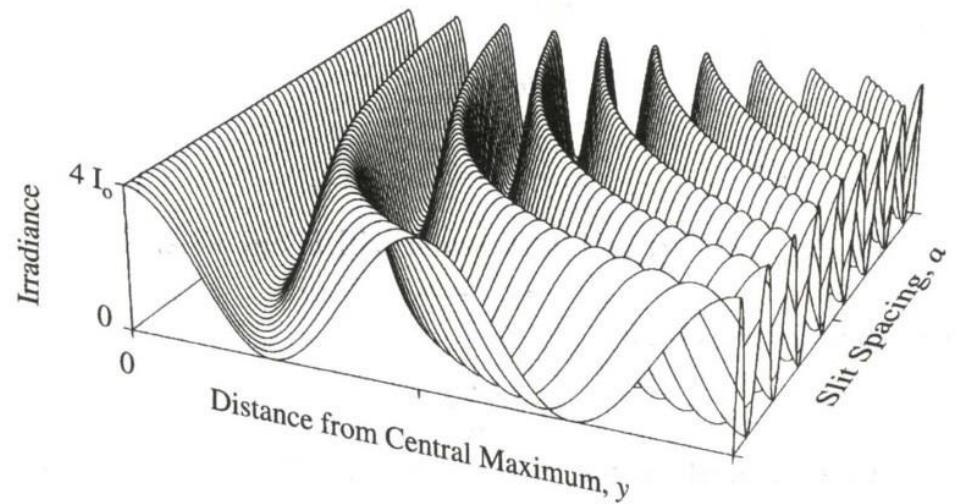
$$r_1 - r_2 = a \frac{y}{s}$$

The resultant irradiance:

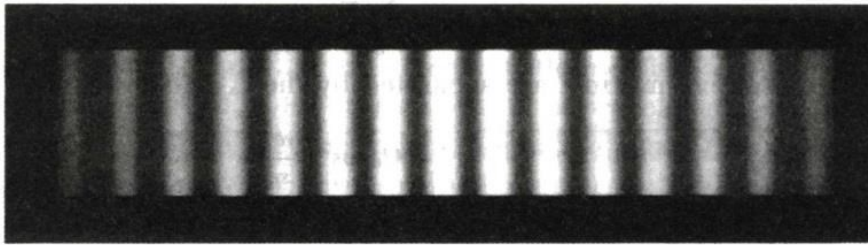
$$I = 4I_0 \cos^2 \frac{ya\pi}{s\lambda}$$



(a)



(b)



Young's double slit experiment

## The effect of finite coherent length

Consider the point further above or below the axis, the  $S_1B$  will increase.

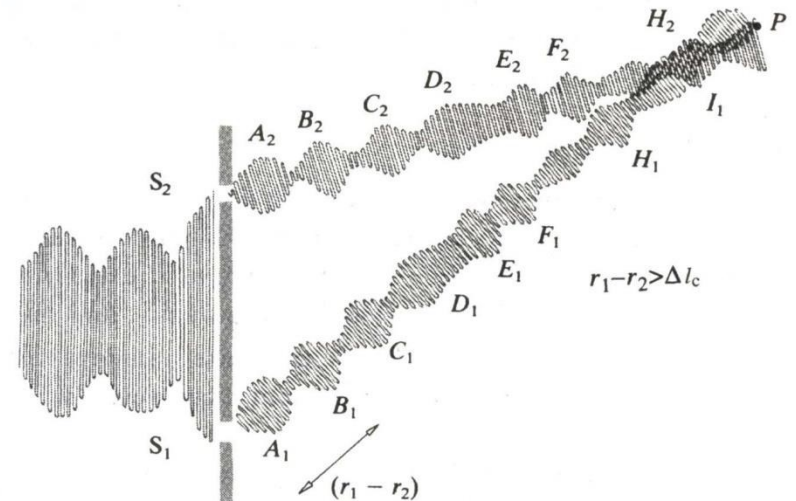
Short coherence length:

- ❑ Identically paired wavegroups will not arrive at P exactly together: the contrast of the fringes will degrade.

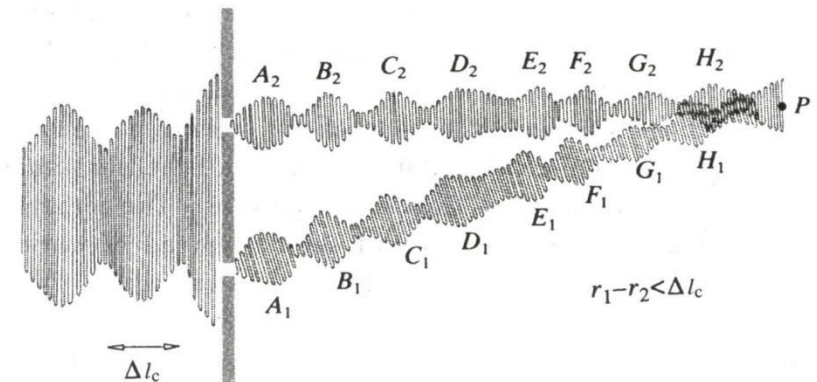
White light:

- ❑ All the constituent colors will arrive at  $y=0$  in phase, so the zeroth-order fringe will be essentially white.
- ❑ All other higher order maxima will show a spread of wavelengths, since

$$y_m = \frac{s}{a} m \lambda.$$



(a)

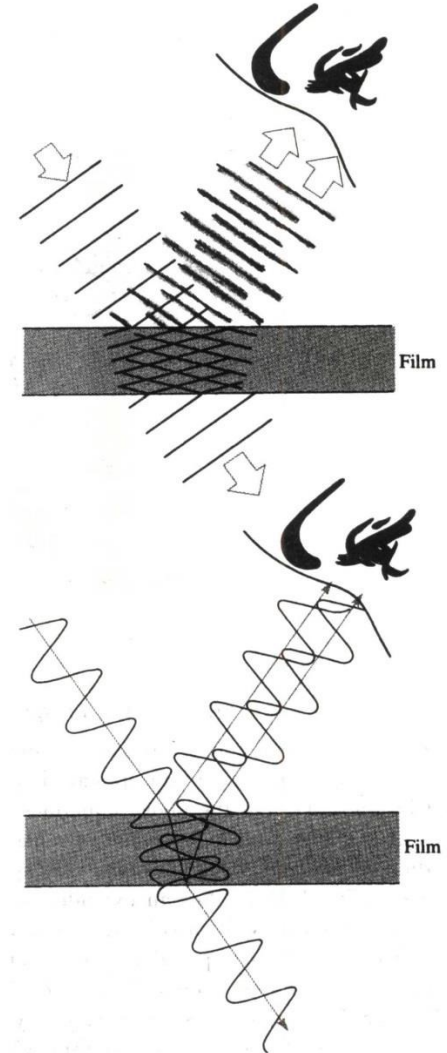


(b)

## 8.4 Amplitude-splitting interferometers

Suppose: a lightwave is incident on a half-silvered mirror or simply on a sheet of glass

- ☐ Part of the wave is transmitted and part reflected.
- ☐ Both the transmitted and reflected waves have lower amplitudes than the original one.
- ☐ The amplitude has been “split”.



## Fringes of equal inclination:

Suppose:

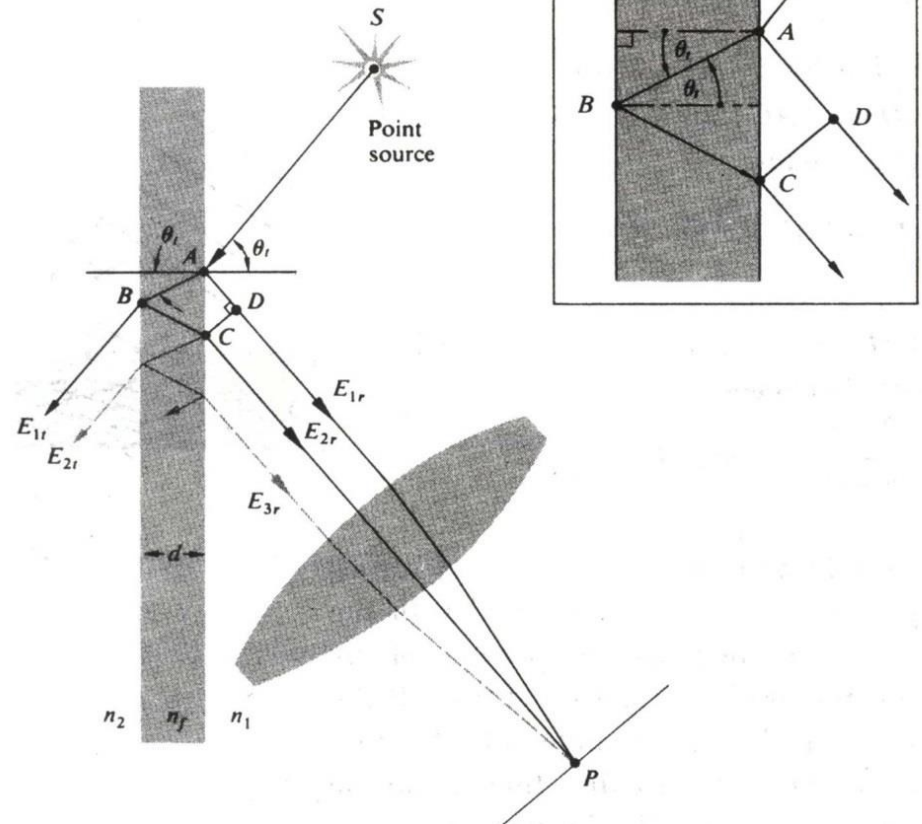
- ❑ the film is nonabsorbing;
- ❑ the amplitude-reflection coefficient at the interfaces are low enough, so only the first two reflected beams are considered.

Optical path-length difference:

$$\Lambda = n_f[(\overline{AB}) + \overline{BC}] - n_1 \overline{AD}$$

Solution:

$$\Lambda = 2n_f d \cos \theta_t$$



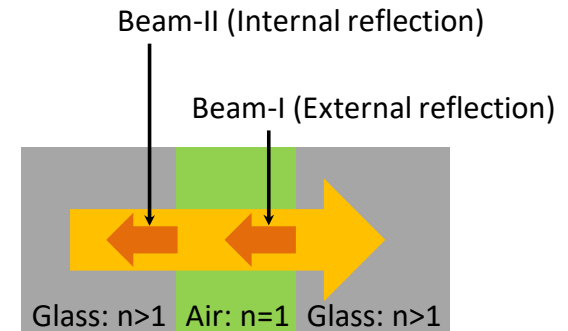
Case I:  $n_1 = n_2 = n; n_f > n$  or  $n_f < n$

A relative  $\pi$  phase shift

The phase difference:

$$\delta = k_0 \Lambda \pm \pi \xrightarrow{\text{for simplicity}} k_0 \Lambda - \pi = \frac{4\pi n_f d \cos \theta_t}{\lambda_0} - \pi$$

A relative  $\pi$  phase shift



When  $\delta = 2m\pi$ , an interference maxima:

$$d \cos \theta_t = \frac{\lambda_f}{4} (2m + 1)$$

When  $\delta = (2m \pm 1)\pi$ , an interference minima:

$$d \cos \theta_t = \frac{\lambda_f}{4} 2m$$

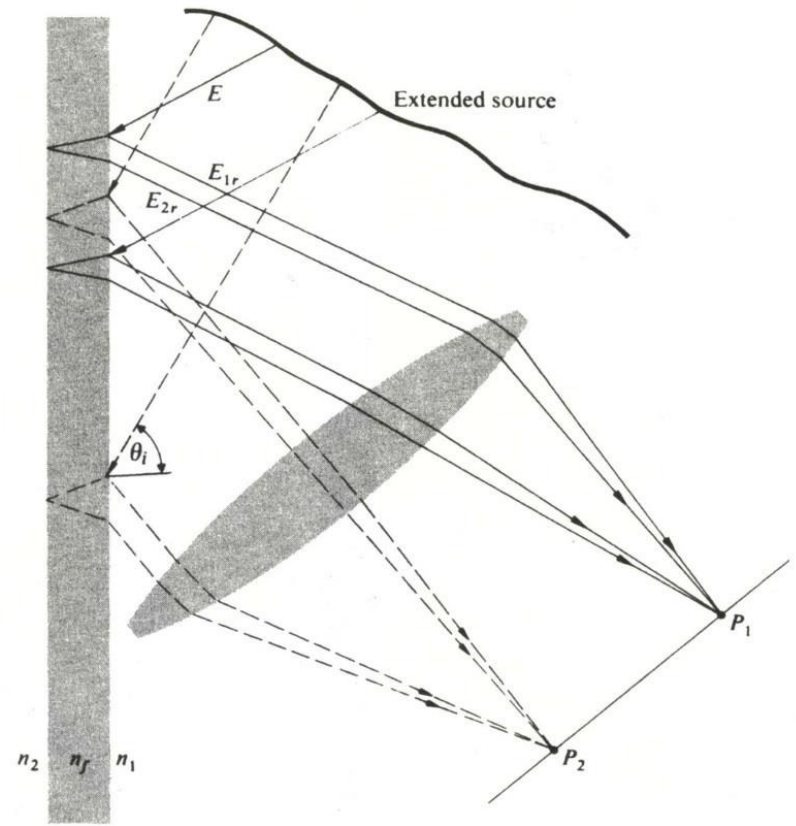
Case II:  $n_1 > n_f > n_2$ ; or  $n_1 < n_f < n_2$

no  $\pi$  phase difference.

## Fringes of equal inclination

All rays inclined at the same angle arrive at the same point

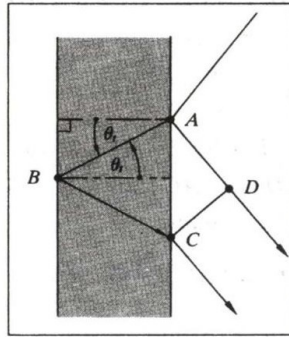
- ❑ The angle  $\theta_i$  or equivalently  $\theta_t$ , determined by the position, will in turn control the phase  $\delta$ .
- ❑ The fringes appearing at points  $P_1$  and  $P_2$ : fringes of equal inclination.
- ❑ Note: each source point on the extended source is incoherent.





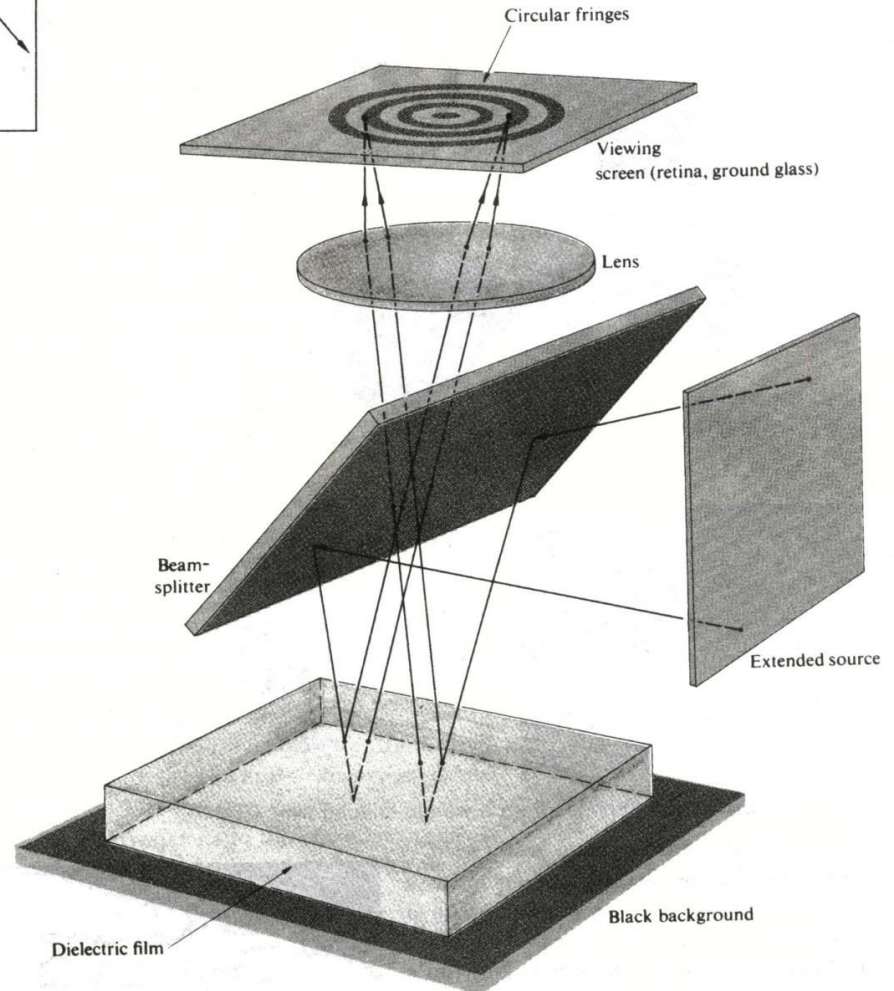
The separation  $\overline{AC}$  between  $E_{1r}$  and  $E_{2r}$ :

$$\overline{AC} = 2d \tan \theta_t$$



As  $d$  increases, the separation of two rays increases. So a larger lens will be used to gather in both rays.

By reducing  $\theta_t$  and therefore  $\theta_i$ , the separation can be reduced, which is known as **Haidinger fringes**.



# Homework

□ 9.1, 9.2, 9.4, 9.26, 9.29



## Fringes of equal inclination

Optical path-length difference:  $\Lambda = 2n_f d \cos \theta_t$

$\theta_i$  ( $\theta_t$ ) is dominant.

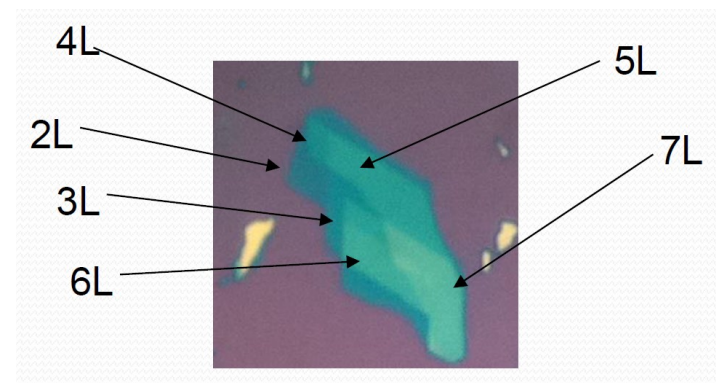
## Fringes of equal thickness

Optical path-length difference:  $n_f d$  is dominant parameter.

Examples:



Iridescence of soap bubbles



Layered MoS<sub>2</sub> materials

## Fizeau fringes (斐佐干涉条纹):

Condition: nearly normal incidence; a thin wedge of small angle  $\alpha$ ;

The optical path length difference:

$$\Lambda = 2n_f d \cos \theta_t$$

$d$  : the thickness at a particular point ( $d = x\alpha$ )

In this case:  $\theta_t \approx \theta_i \approx 0$ ,  $\Lambda = 2n_f d$

An interference maximum (including a relative phase shift of  $\pi$  radians)

$$(m + 1/2)\lambda_0 = 2n_f d_m$$

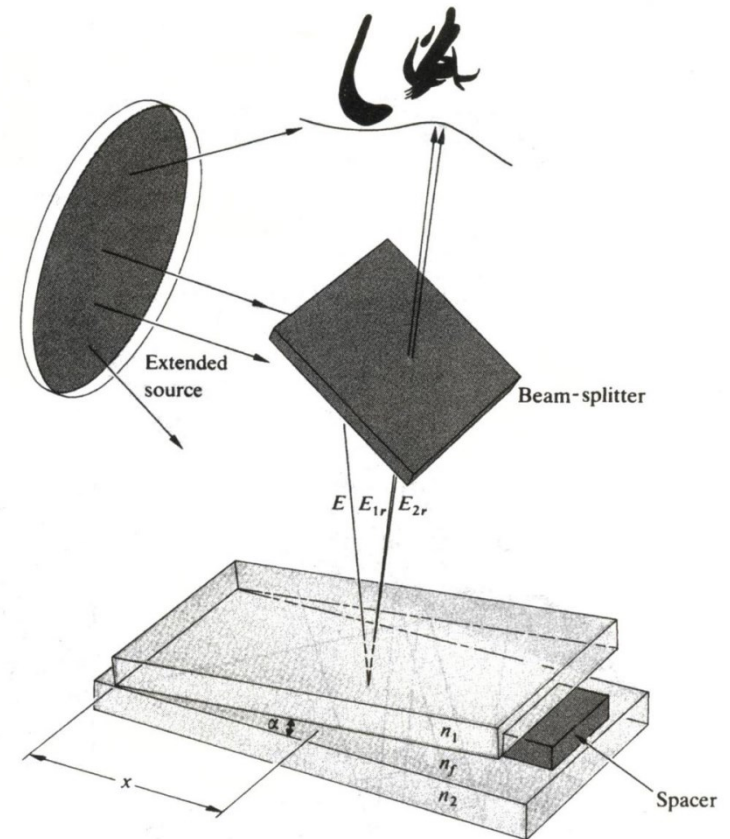
$$d_m = \left(m + \frac{1}{2}\right) \frac{\lambda_f}{2}$$

Maxima occurs at an odd multiple of a quarter wavelength.

$$(m + 1/2)\lambda_0 = 2n_f \alpha x_m$$

$$x_m = \left(\frac{m + 1/2}{2\alpha}\right) \lambda_f$$

Maxima occur at distances from the apex given by  $\lambda_f/2\alpha$ ,  $3\lambda_f/4\alpha$ , and so on.



Monochromatic fringes

# Example I: a wedge-shaped film made of liquid dishwashing soap

## The black region at the top:

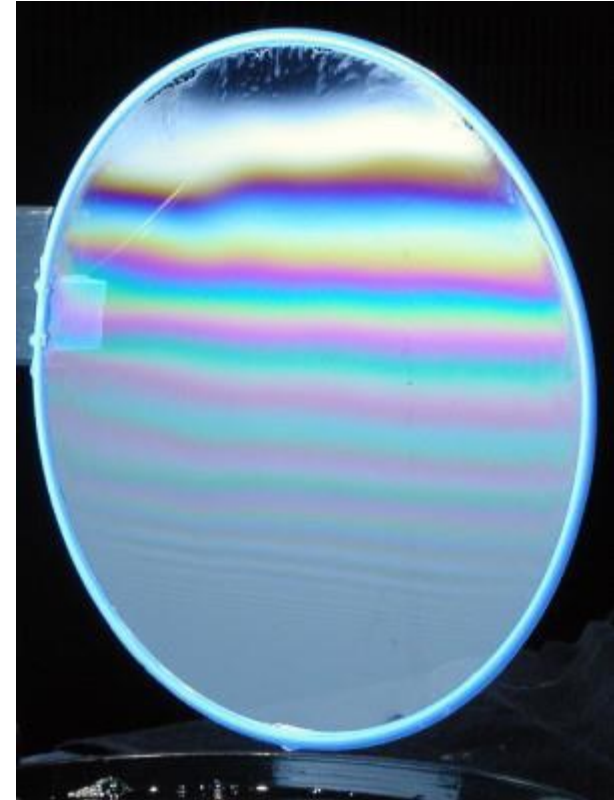
- ❑ The film thickness is less than  $\lambda_f/4$ ; the reflected rays from two surfaces are  $\lambda_f/2$  phase shift; the reflected rays are out of phase.

## The soap film increases in thickness towards the bottom:

- ❑ Reflections from the front and rear of the soap film gradually undergo increasingly larger phase shifts.
- ❑ For example, when the thickness is  $1/4$  wavelength, the two reflections is in phase and thereby giving a maximum reflected light intensity.
- ❑ As we continue to increase in thickness, every  $1/4$  wavelength increase in thickness we move from a region of no reflection to a region of maximum reflection to no reflection again, etc.

## The spacing of the light and dark reflection bands across the soap film:

- ❑ An indication of how the thickness varies.



## Example II: Newton's ring

Condition: a lens is placed on an optical flat and illuminated at normal incidence with quasimonochromatic light.

The distance  $x$ :

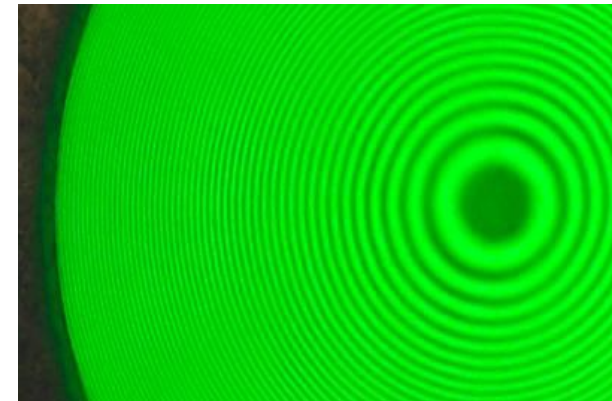
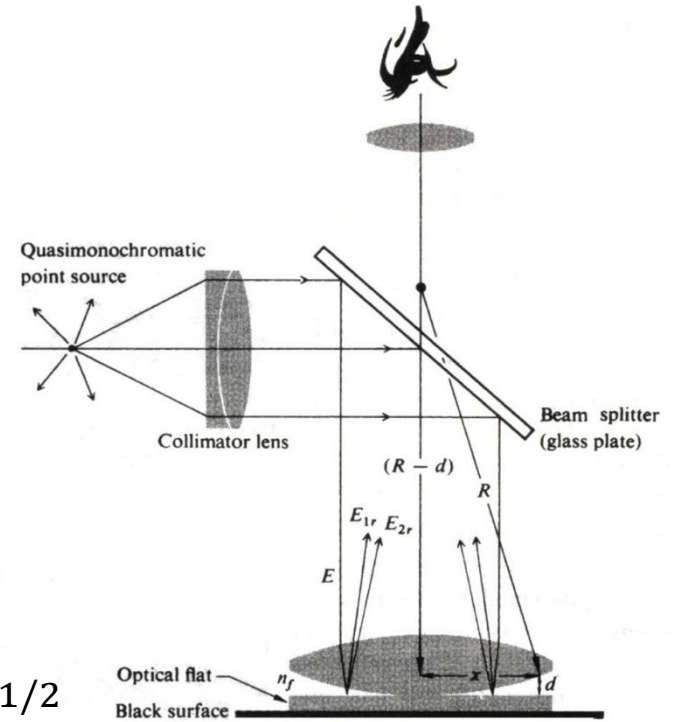
$$x^2 = R^2 - (R - d)^2 = 2Rd - d^2 \xrightarrow{R \gg d} 2Rd$$

The  $m$ th-order interference maximum occurs at:

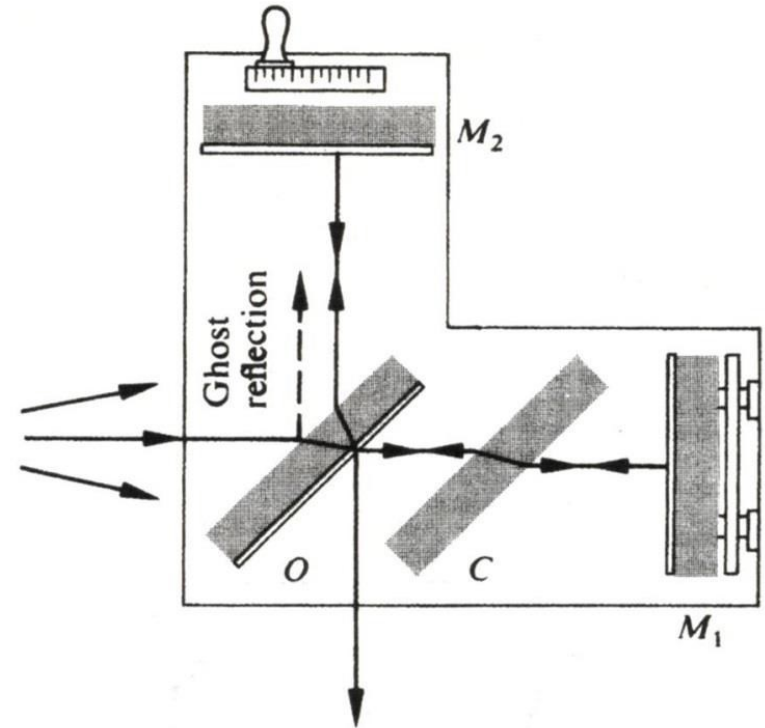
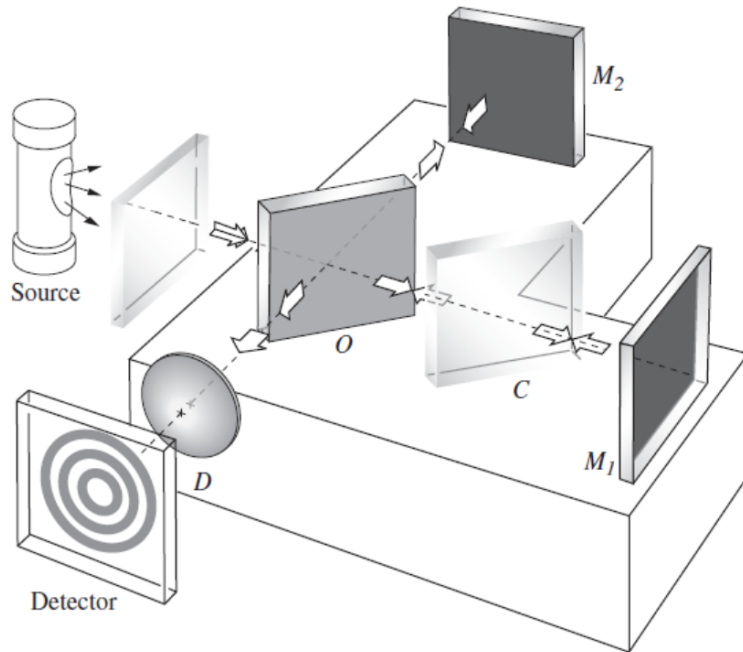
$$2n_f d_m = \left(m + \frac{1}{2}\right) \lambda_0 \Rightarrow x_m = \left[ \left(m + \frac{1}{2}\right) \lambda_f R \right]^{1/2}$$

The  $m$ th-order interference minimum occurs at:

$$2n_f d_m = m \lambda_0 \Rightarrow x_m = (m \lambda_f R)^{1/2}$$



# Michelson Interferometer



The compensator (C): an exact duplicate of the beamsplitter.

Function:

- ❑ Compensate the difference between any optical path and actual path.
- ❑ Negate the effect of dispersion, so a source with a very broad bandwidth will generate discernible fringes.

# How Michelson interferometer works

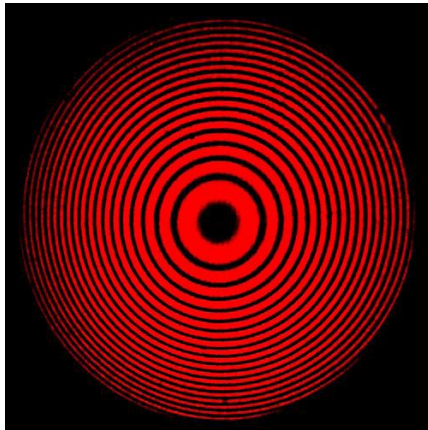
Fringes of equal inclination:

Destructive interference will exist (consider the relative phase shift of  $\pi$  radians) when

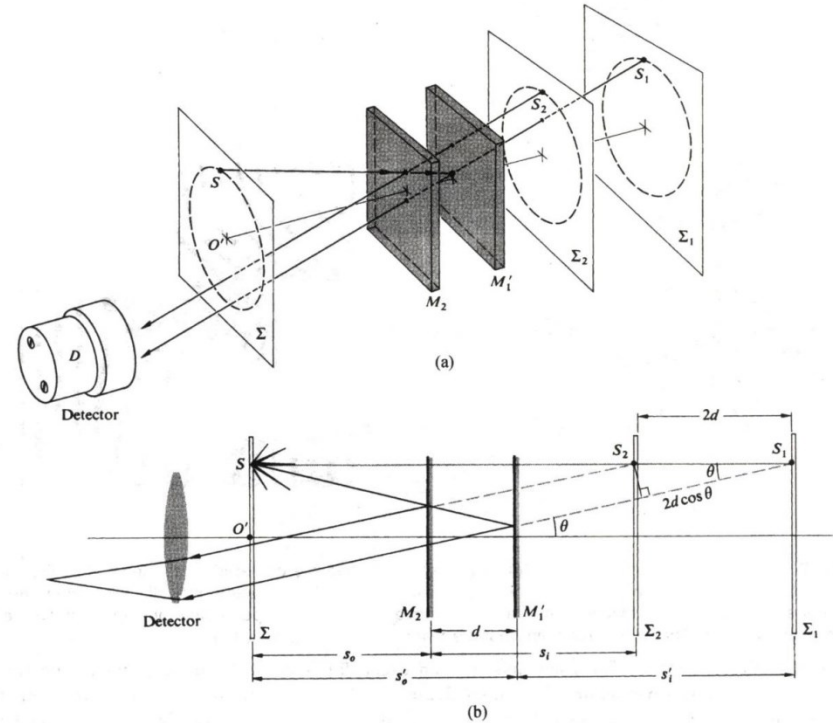
$$2d \cos \theta_m = m\lambda_0$$

The interference pattern:

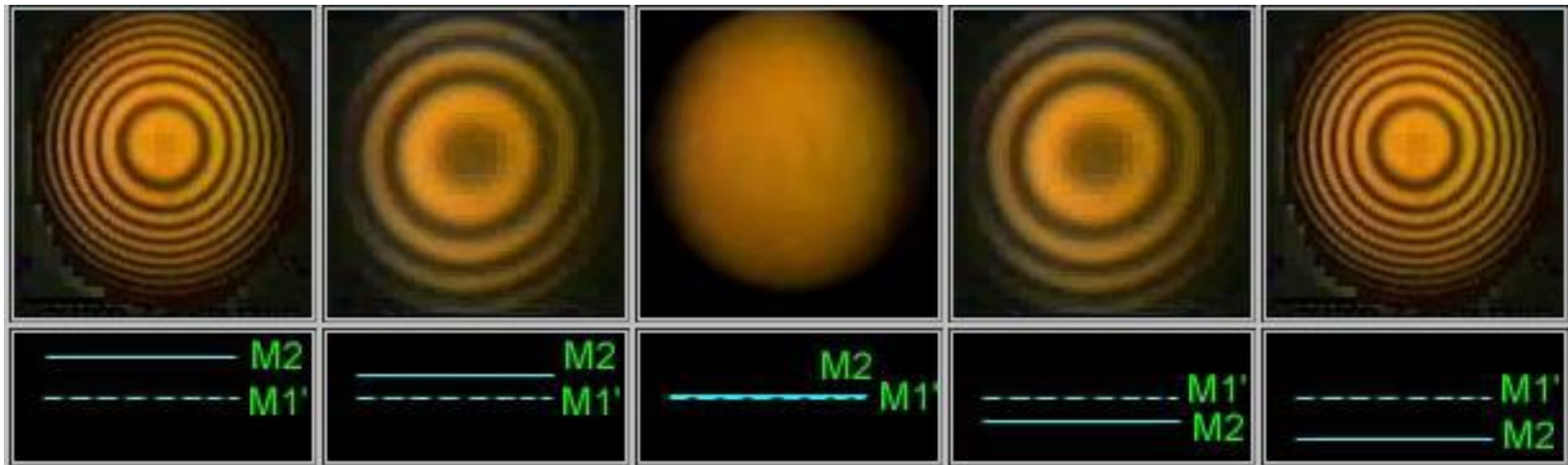
A circular fringe system concentric with the central axis.



Note:  $2d \cos \theta_m$  must be less than the coherence length of the source, so laser light will be easier than a tungsten bulb or a candle to use in demonstrating the interferometer.



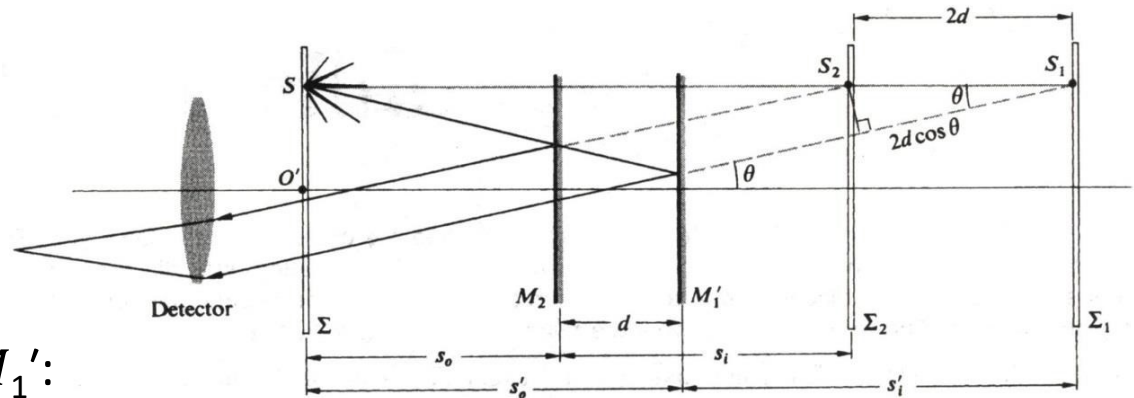




$$2d \cos \theta_m = m\lambda_0$$

As  $M_2$  is moved toward  $M_1'$ :  
 $d$  decreases,  
 $\cos \theta_m$  increases while  $\theta_m$  decreases.

Phenomenon: the rings shrink toward the center; each remaining ring broadens as more and more fringes vanish at the center; by  $d=0$ , the central fringe will have spread out, filling the entire field of view.



A central dark fringe  $\theta_m = 0$

$$2d = m_0 \lambda_0$$

Set  $d = 10 \text{ cm}$ ;  $\lambda_0 = 500 \text{ nm}$ ;

$$m_0 = \frac{2d}{\lambda_0} = 400000$$

Successive dark rings:

$$2d \cos \theta_1 = (m_0 - 1) \lambda_0$$

$$2d \cos \theta_2 = (m_0 - 2) \lambda_0$$

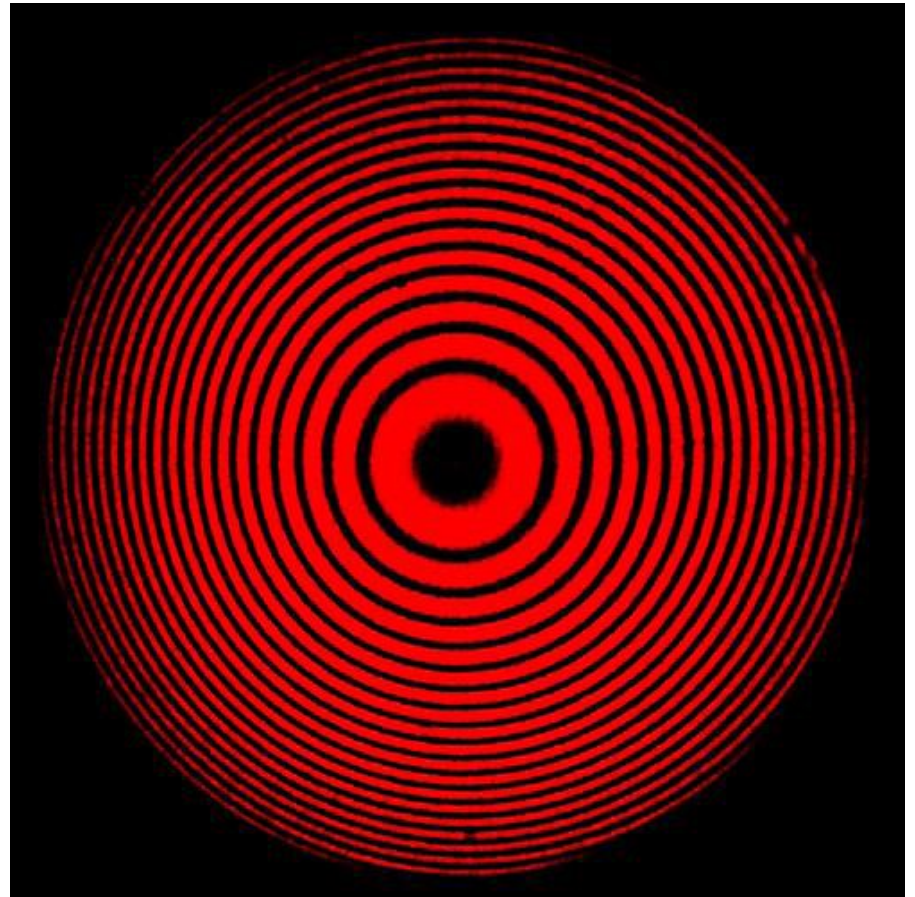
$\vdots$

$$2d \cos \theta_p = (m_0 - p) \lambda_0$$

For the p-th ring:

$$2d(1 - \cos \theta_p) = p \lambda_0$$

The sixth dark ring can be specified by stating that  $p=6$ , corresponding to  $m=399994$  in the above case.





## Wavelength measurement

As the movable mirror is displaced by  $\lambda_0/2$ , each fringe will move to the position previously occupied by an adjacent fringe.

$$\Delta d = N \left( \frac{\lambda_0}{2} \right)$$

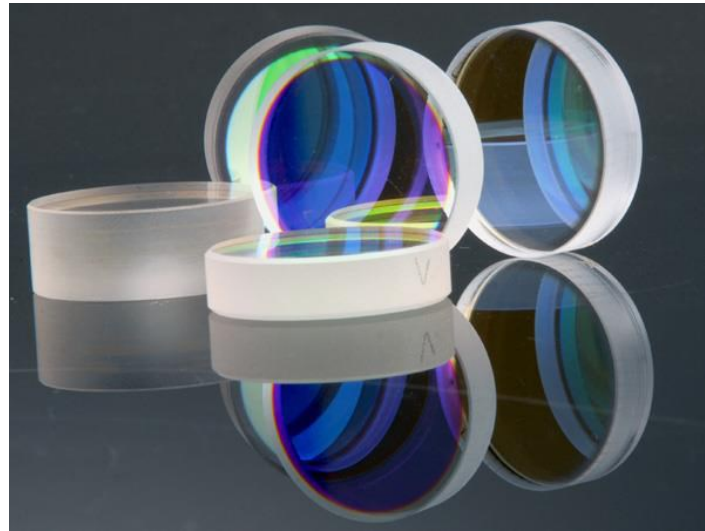
Nowadays this can be done fairly easily by electronic means:  
**spectrometer**

# Application of single and multilayer films

Coating to eliminate unwanted reflections off a diversity of surfaces, from showcase glass to high-quality camera lenses, are now commonplace.



Coated camera lenses



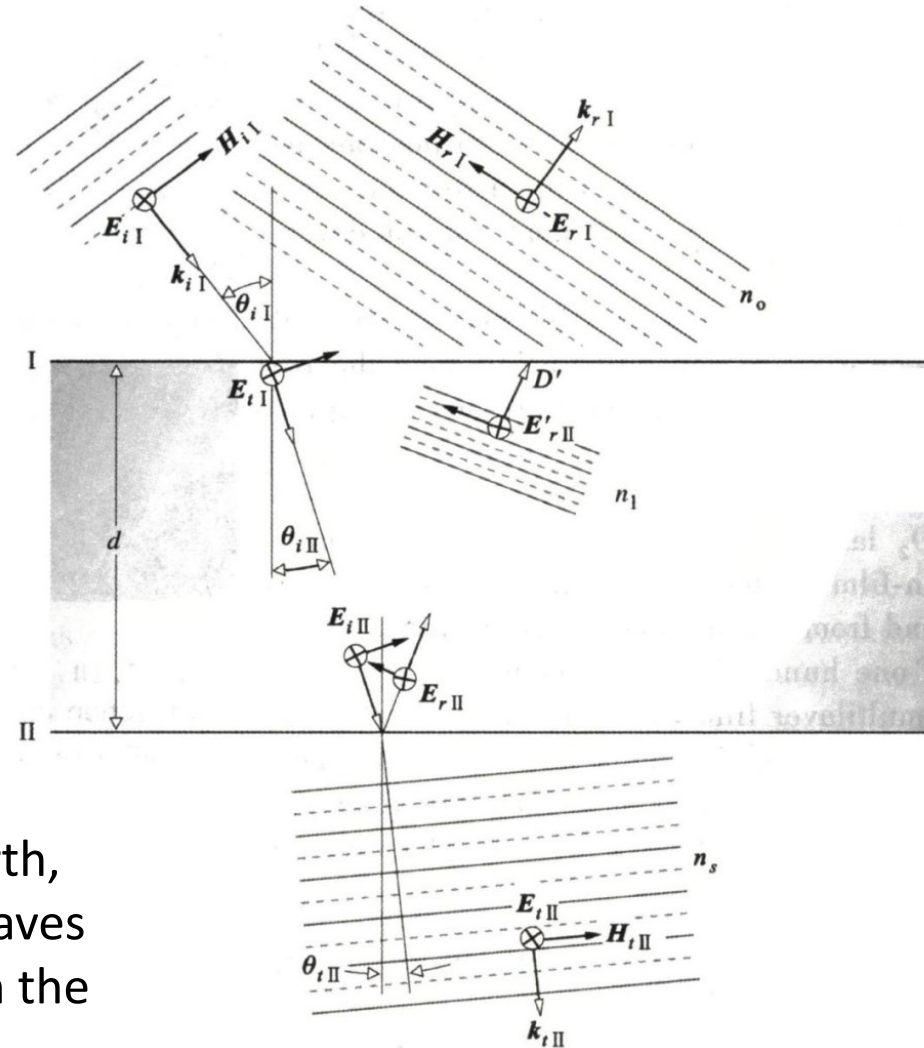
Dichroic mirrors

## Mathematical Treatment

Consider the linearly polarized wave impinging on a thin dielectric film between two semi-infinite transparent media.

Boundary conditions: the tangential components of both the electric ( $\vec{E}$ ) and magnetic field ( $\vec{H} = \frac{\vec{B}}{\mu} = \sqrt{\frac{\epsilon_0}{\mu_0}} n \vec{k} \times \vec{E}$ ) be continuous across the boundaries.

Note: each wave  $E_{rI}$ ,  $E_{rII}'$ ,  $E_{tII}$ , and so forth, represents the resultant of all possible waves traveling in that direction, at that point in the medium.



At boundary I:

$$E_I = E_{iI} + E_{rI} = E_{tI} + E_{rII}'$$

$$H_I = \sqrt{\frac{\epsilon_0}{\mu_0}} (E_{iI} - E_{rI}) n_0 \cos \theta_{iI}$$

$$= \sqrt{\frac{\epsilon_0}{\mu_0}} (E_{tI} - E_{rII}') n_1 \cos \theta_{iII}$$

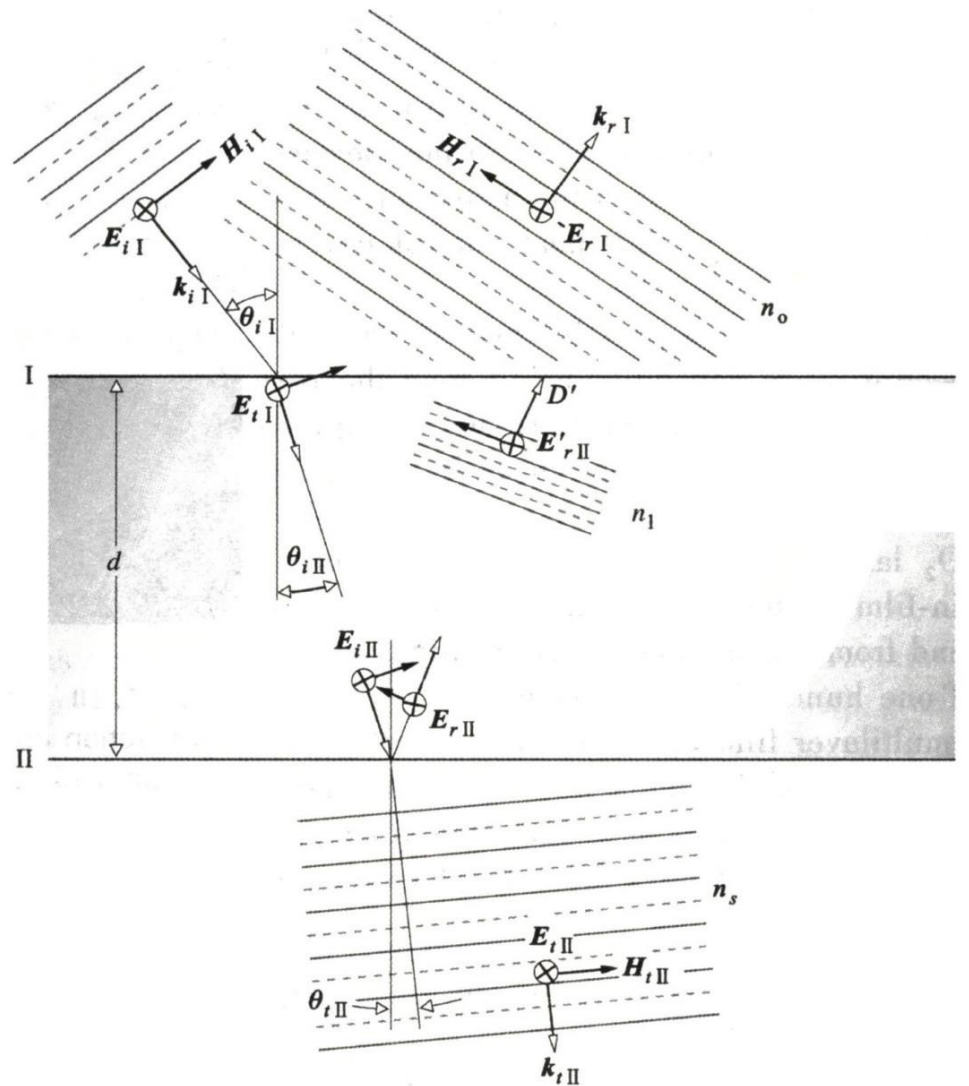
At boundary II:

$$E_{II} = E_{iII} + E_{rII} = E_{tII}$$

$$H_{II} = \sqrt{\frac{\epsilon_0}{\mu_0}} (E_{iII} - E_{rII}) n_1 \cos \theta_{iII}$$

$$= \sqrt{\frac{\epsilon_0}{\mu_0}} E_{tII} n_s \cos \theta_{tII}$$

$$\begin{bmatrix} E_I \\ H_I \end{bmatrix} = \begin{bmatrix} \cos k_0 h & \frac{i \sin k_0 h}{\gamma_1} \\ \gamma_1 i \sin k_0 h & \cos k_0 h \end{bmatrix} \begin{bmatrix} E_{II} \\ H_{II} \end{bmatrix}$$



$$k_0 h = k_0 (2n_1 d \cos \theta_{iII}) / 2$$

$$\gamma_1 = \sqrt{\frac{\epsilon_0}{\mu_0}} n_1 \cos \theta_{iII}$$

$$\begin{bmatrix} E_I \\ H_I \end{bmatrix} = \mathcal{M}_I \begin{bmatrix} E_{II} \\ H_{II} \end{bmatrix}$$

$$\mathcal{M}_I = \begin{bmatrix} \cos k_0 h & \frac{i \sin k_0 h}{\gamma_1} \\ \gamma_1 i \sin k_0 h & \cos k_0 h \end{bmatrix}$$

The characteristic matrix  $\mathcal{M}_I$  relates the fields at the two adjacent boundaries.

For two overlaying films deposited on the substrate

$$\begin{bmatrix} E_{II} \\ H_{II} \end{bmatrix} = \mathcal{M}_{II} \begin{bmatrix} E_{III} \\ H_{III} \end{bmatrix}$$

For  $p$  layers, each with a particular value of  $n$  and  $h$ :

$$\begin{bmatrix} E_I \\ H_I \end{bmatrix} = \mathcal{M}_I \mathcal{M}_{II} \dots \mathcal{M}_p \begin{bmatrix} E_{(p+1)} \\ H_{(p+1)} \end{bmatrix}$$

$$\mathcal{M} = \mathcal{M}_I \mathcal{M}_{II} \dots \mathcal{M}_p = \begin{bmatrix} m_{11} & m_{12} \\ m_{21} & m_{22} \end{bmatrix}$$

At boundary I:

$$E_I = E_{iI} + E_{rI} = E_{tI} + E_{rII}'$$

$$H_I = \gamma_0(E_{iI} - E_{rI}) = \gamma_1(E_{tI} - E_{rII}')$$

At boundary II:

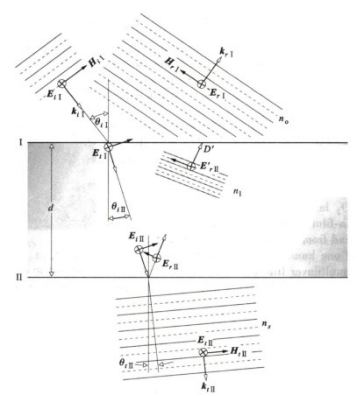
$$E_{II} = E_{iII} + E_{rII} = E_{tII}$$

$$H_{II} = \gamma_1(E_{iII} - E_{rII}) = \gamma_s E_{tII}$$

$$\gamma_0 \equiv \sqrt{\frac{\epsilon_0}{\mu_0}} n_0 \cos \theta_{iI}$$

$$\gamma_1 \equiv \sqrt{\frac{\epsilon_0}{\mu_0}} n_1 \cos \theta_{iII}$$

$$\gamma_s \equiv \sqrt{\frac{\epsilon_0}{\mu_0}} n_s \cos \theta_{tII}$$



$$\begin{bmatrix} E_I \\ H_I \end{bmatrix} = \mathcal{M}_I \begin{bmatrix} E_{II} \\ H_{II} \end{bmatrix} \quad \Rightarrow \quad \begin{bmatrix} (E_{iI} + E_{rI}) \\ (\gamma_0(E_{iI} - E_{rI})) \end{bmatrix} = \mathcal{M}_I \begin{bmatrix} E_{tII} \\ \gamma_s E_{tII} \end{bmatrix}$$

$$\Rightarrow \begin{cases} 1 + r = m_{11}t + m_{12}\gamma_s t \\ (1 - r)\gamma_0 = m_{21}t + m_{22}\gamma_s t \end{cases}$$

The amplitude coefficients of reflection and transmission

$$\Rightarrow \begin{cases} r = \frac{\gamma_0 m_{11} + \gamma_0 \gamma_s m_{12} - m_{21} - \gamma_s m_{22}}{\gamma_0 m_{11} + \gamma_0 \gamma_s m_{12} + m_{21} + \gamma_s m_{22}} \\ t = \frac{2\gamma_0}{\gamma_0 m_{11} + \gamma_0 \gamma_s m_{12} + m_{21} + \gamma_s m_{22}} \end{cases}$$

## Antireflection (增透) coatings

Consider the important case of normal incidence:  $\theta_{iI} = \theta_{iII} = \theta_{tII} = 0$

The reflection coefficient for a single film:

$$r_1 = \frac{n_1(n_0 - n_s) \cos k_0 h + i(n_0 n_s - n_1^2) \sin k_0 h}{n_1(n_0 + n_s) \cos k_0 h + i(n_0 n_s + n_1^2) \sin k_0 h}$$

$$R_1 = \frac{n_1^2 (n_0 - n_s)^2 \cos^2 k_0 h + (n_0 n_s - n_1^2)^2 \sin^2 k_0 h}{n_1^2 (n_0 + n_s)^2 \cos^2 k_0 h + (n_0 n_s + n_1^2)^2 \sin^2 k_0 h}$$

When  $k_0 h = \frac{\pi}{2}$  (equivalent to  $d = \frac{\lambda_f}{4}$ )

$$R_1 = \frac{(n_0 n_s - n_1^2)^2}{(n_0 n_s + n_1^2)^2} \quad \text{which equals to zero, when } n_0 n_s = n_1^2$$

Example:

Cryolite ( $n=1.35$ ) or  $\text{MgF}_2$  ( $n=1.38$ ) on glass ( $n=1.5$ ): the reflectance of glass will reduce from 4% to 1%.



## A double-layer, **quarter-wavelength** antireflection coating

$$k_0 h = \frac{\pi}{2} \Rightarrow \mathcal{M} = \mathcal{M}_I \mathcal{M}_{II} = \begin{bmatrix} 0 & i/\gamma_1 \\ i\gamma_1 & 0 \end{bmatrix} \begin{bmatrix} 0 & i/\gamma_2 \\ i\gamma_2 & 0 \end{bmatrix}$$

At normal incidence:

$$\mathcal{M} = \begin{bmatrix} -n_2/n_1 & 0 \\ 0 & -n_1/n_2 \end{bmatrix}$$

Reflectance:

$$R_2 = \left[ \frac{n_2^2 n_0 - n_s n_1^2}{n_2^2 n_0 + n_s n_1^2} \right]^2$$

For  $R_2 = 0$  at a particular wavelength:  $\left( \frac{n_2}{n_1} \right)^2 = \frac{n_s}{n_0}$

Since  $n_2 > n_1$ ; it is common practice to designate a (glass)-(high index)-(low index)-(air) system as gHLA.

Example:  $\text{ZrO}_2$  ( $n=2.1$ );  $\text{TiO}_2$  ( $n=2.4$ );  $\text{ZnS}$  ( $n=2.32$ ) are used for H-layer;  $\text{MgF}_2$  ( $n=1.38$ );  $\text{CeF}_3$  (Cerium Fluoride,  $n=1.63$ ) are employed as L-layer.

**High reflectance:** if  $n_2/n_1 \gg 1$ ,  $R = 1$

To further enhance the reflection, one can use a periodic quarter-wave stack system  $g(\text{HL})^m a$

$$\mathcal{M} = \mathcal{M}_L \mathcal{M}_H \mathcal{M}_L \mathcal{M}_H \dots \mathcal{M}_L \mathcal{M}_H$$

$$\mathcal{M}_L \mathcal{M}_H = \begin{bmatrix} 0 & i/\gamma_L \\ i\gamma_L & 0 \end{bmatrix} \begin{bmatrix} 0 & i/\gamma_H \\ i\gamma_H & 0 \end{bmatrix} = \begin{bmatrix} -n_H/n_L & 0 \\ 0 & -n_L/n_H \end{bmatrix}$$

$$\mathcal{M} = \begin{bmatrix} (-n_H/n_L)^m & 0 \\ 0 & (-n_L/n_H)^m \end{bmatrix}$$

$$R = \left[ \frac{n_0(n_H)^{2m} - n_s(n_L)^{2m}}{n_0(n_H)^{2m} + n_s(n_L)^{2m}} \right]^2$$

When  $n_H/n_L \gg 1$  and  $m$  is large the reflectance can be very high.

More improvement can be achieved by adding one more high-index layer to the air side of the periodic system to become  $g(\text{HL})^m \text{Ha}$  system.

# Homework

□ 9.31, 9.33, 9.45, 9.46